

Midterm Review: Sessions 1–8 Synthesis

Session 9 • Unit 2 • Preparation for the midterm exam

Samir Asaf, PhD, CFA, CMA, CTP, CM&AA

Senior Partner, Regent Financial LLC, New York

Former Finance Instructor, Stanford University

Liquidity Illusion: The General Equilibrium Theory of Private Capital Valuation (Forthcoming, 2026)

Graduate Finance Course • Session 9 of 32 • Spring 2027

Today: integration and exam preparation

1

The conceptual arc

Diagnosis → measurement → requirements

2

Key equations review

DCF, LAV, Jensen, OU, LA-PME

3

Calibrated numbers

The 25 numbers you should know

4

Question type walkthroughs

Multiple-choice, problem, case

5

Formula sheet construction

How to organize your one-page reference

The conceptual arc of Unit 1 + Unit 2

S1-2	Why DCF fails	Three structural failures • scale of stakes
S3	Empirical evidence	Secondary market 2003-2024 • four regimes
S4	OU process	Mathematical model of L_t dynamics
S5	Term structure	Jensen bias $B(T) \approx A \cdot T$ • DLOM comparison
S6	IRR biases	Reinvestment • sign-pattern
S7	PME and LA-PME	Public benchmarking • liquidity adjustment
S8	Five requirements	DCF/DLOM/factors fail • only GE-LAV passes

The five equations to memorize

DCF	$V_0 = \sum_{t=1}^T \frac{CF_t}{(1+r)^t}$	Constant discount rate, deterministic
OU SDE for L_t	$dL_t = \kappa(\bar{L} - L_t) dt + \sigma dW_t$	Stochastic mean-reverting process • $\kappa=0.45$, $\sigma=0.32$, $\bar{L}=1.0$
Quadratic premium	$\pi(L) = 0.045 - 0.025L + 0.021L^2$	Convex in L • always positive • matches secondary market
LAV operator	$V^{LAV} = E \left[\sum_t CF_t \cdot e^{-\int_0^t r(L_s) ds} \right]$	Expectation over OU path
Jensen bias	$B(T) \approx A \cdot T, \quad A \approx 0.17\%/yr$	Linear in horizon

The 12 numbers you should know cold

\$13T

Private capital AUM (2025)

\$73MMaster anchor: DCF-GE-LAV gap on 20yr
infra**0.45/yr**OU mean-reversion κ **0.32**OU volatility σ **0.337**Stationary std of L_t **3.5%**Industry-standard illiquidity premium
(wrong)**1.6-4.5%**

Jensen bias range 10-25yr horizons

4.31×Crisis amplification (r_{GE}/r_{DCF} at GFC) **$L = -1.5$** GFC-depth L_t state**2.3%/yr**

Welfare gap (Pigouvian externality)

7%Pigouvian tax τ^* at GFC depth (vs 0% normal)**70.1%**

Mark-down gap DCF→GE-LAV at GFC

Midterm question types (4 sections, 75 min)

Multiple choice

30 min • 10 questions

Concept recognition. Definitions, mechanism identification, what-if framing. Each correctly answers a learning objective.

→ 1.5 points each

Short answer

20 min • 4 questions

Brief explanations (3-5 sentences). 'Why is the Jensen bias linear in horizon?' Type questions.

→ 5 points each

Quant problem

15 min • 1 problem

Compute a specific GE-LAV quantity. E.g., compute LAV value given OU params and a cash flow stream.

→ 15 points

Plus integrative case (10 min, 1 short scenario) — 20 points. Total: 100 points.

How to build your formula sheet (8.5×11, handwritten, both sides)

Front: Equations

1. DCF formula
2. OU SDE + parameter values
3. Quadratic premium $\pi(L)$
4. LAV operator
5. Jensen bias $B(T) \approx A \cdot T$
6. LA-PME formula
7. Kaplan-Schoar PME

Front: Numbers

$\kappa=0.45/\text{yr} \cdot \sigma=0.32 \cdot \bar{L}=1.0$
 $\sigma_{L\infty} = 0.337$
 $A \approx 0.16\text{-}0.18\%/\text{yr}$
 Master anchor: \$103.6M / \$30.9M / \$73M
 GFC: $L = -1.5$
 Crisis amp: 4.31×

Back: Concepts

1. Three structural failures (named)
2. Five requirements scorecard
3. Four secondary market regimes
4. Level vs convexity illusion
5. IRR's two biases
6. Why all PME variants are constant-rate

Back: Examples

1. Stationary distribution: $N(\bar{L}, \sigma^2/2\kappa)$
2. Discrete OU recursion
3. Worked KS-PME calculation
4. Discount-to-NAV interpretation
5. The Vasicek connection

Tip: writing it forces memorization. Don't print and paste.

Session 9 wrap-up • midterm preparation

Key takeaways

1

The arc: diagnosis (S1-2) → evidence (S3) → math (S4-5) → measurement (S6-7) → requirements (S8).

2

Five equations to memorize: DCF, OU SDE, quadratic premium, LAV operator, Jensen bias.

3

Twelve numbers to know cold: κ , σ , $\bar{L}$, \$13T, \$73M anchor, 4.31× amp factor, etc.

4

Practice writing the formula sheet by hand tonight.

5

Office hours tomorrow 4-6pm for last-minute questions.

NEXT: SESSION 10

Session 10: MIDTERM EXAM (75 min, in-class).

Arrive 10 min early.

Bring: pen, pencil, eraser, scientific calculator, handwritten formula sheet, photo ID.

Coverage: Sessions 1-8 entirely.

Format: 10 MC + 4 short answer + 1 quant + 1 case.

See midterm-blueprint.qmd on the site.